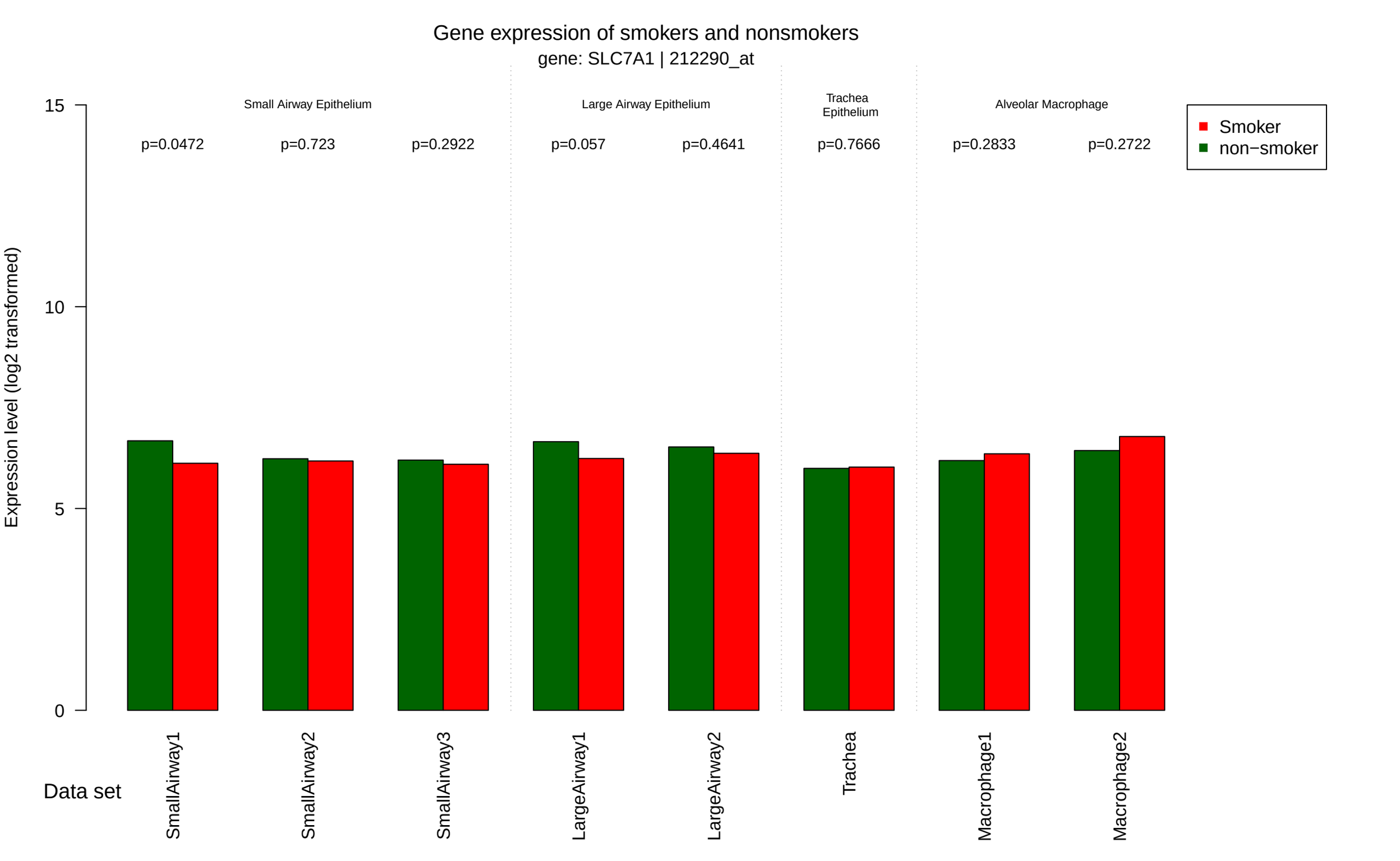
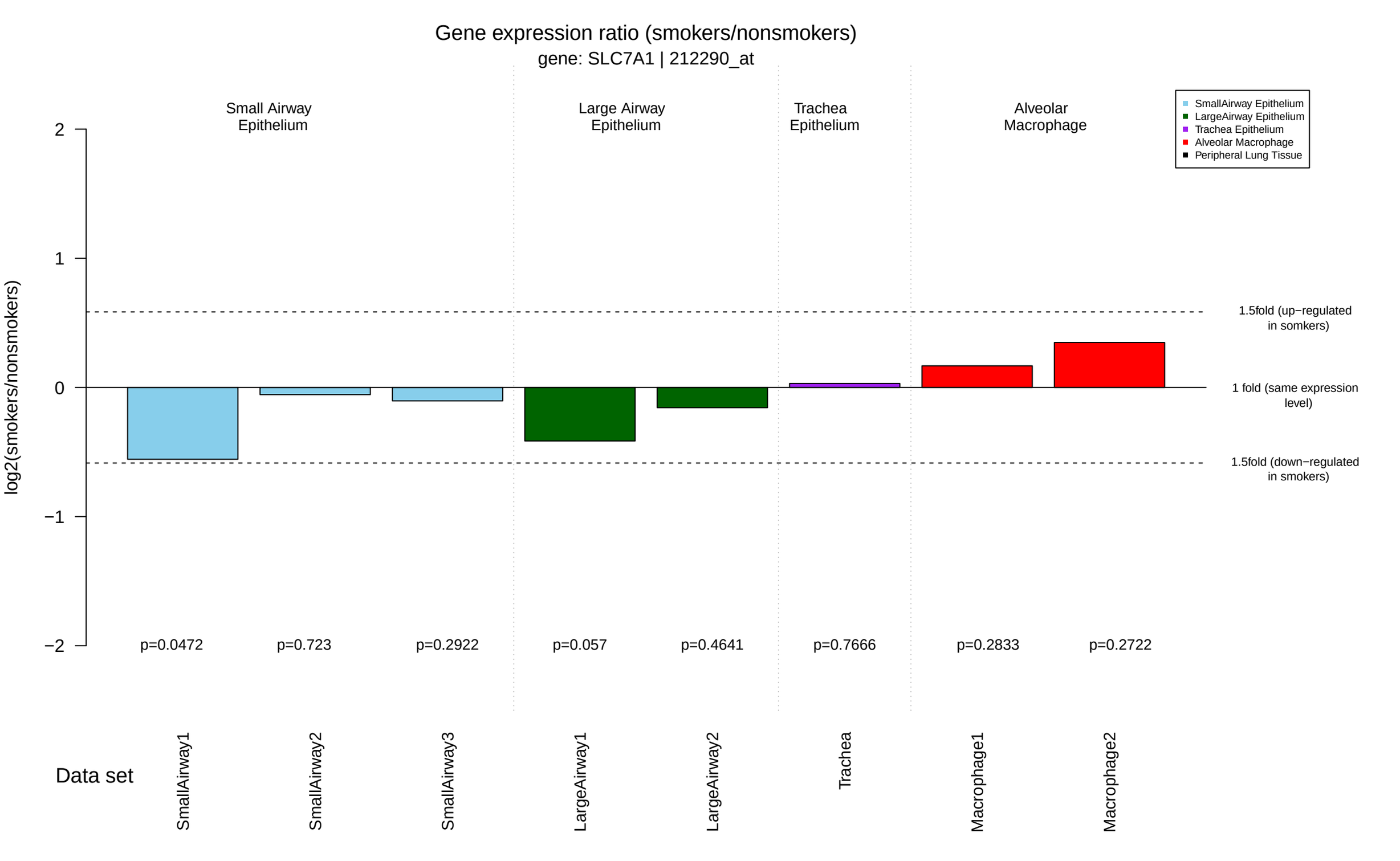
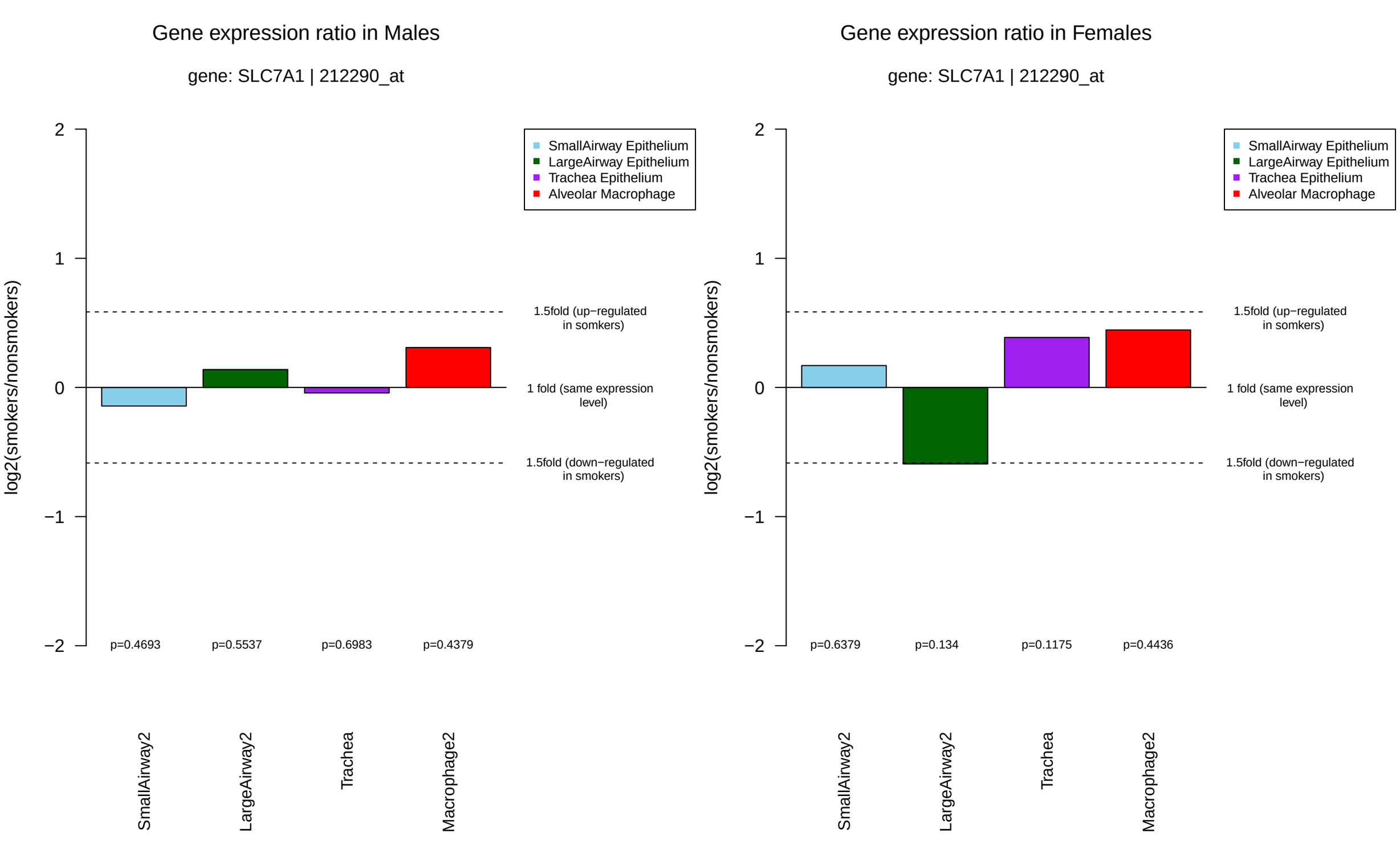
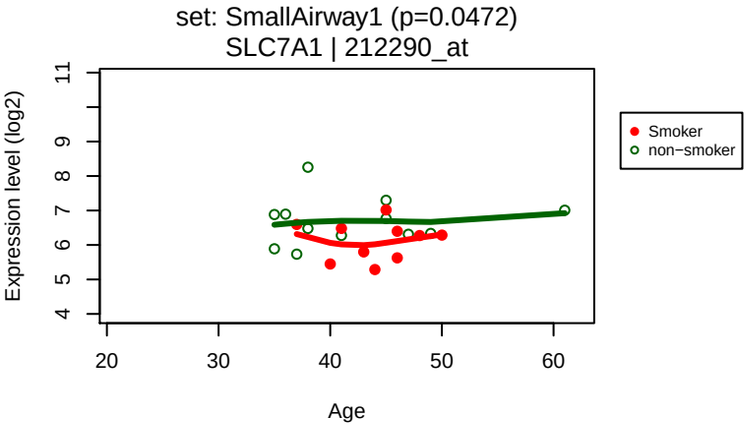
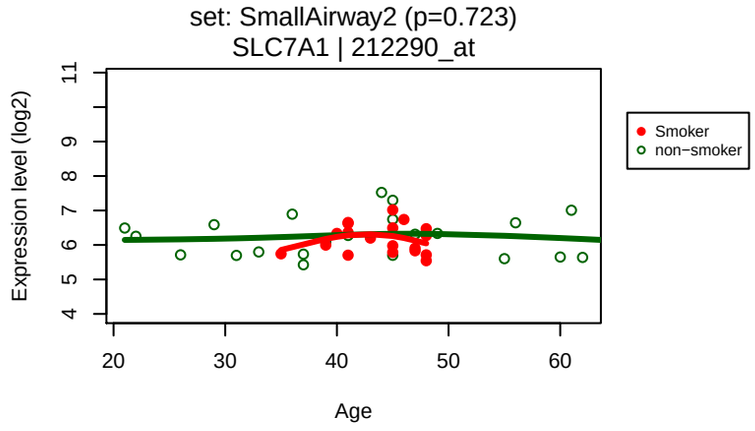
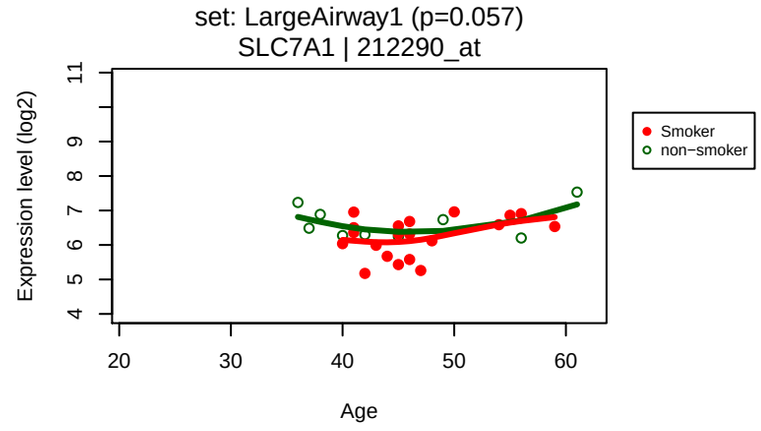

Gene expression of smokers and nonsmokers
gene: SLC7A1 | 212290_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0472
p=0.723
p=0.2922
p=0.057
p=0.4641
p=0.7666
p=0.2833
p=0.2722
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

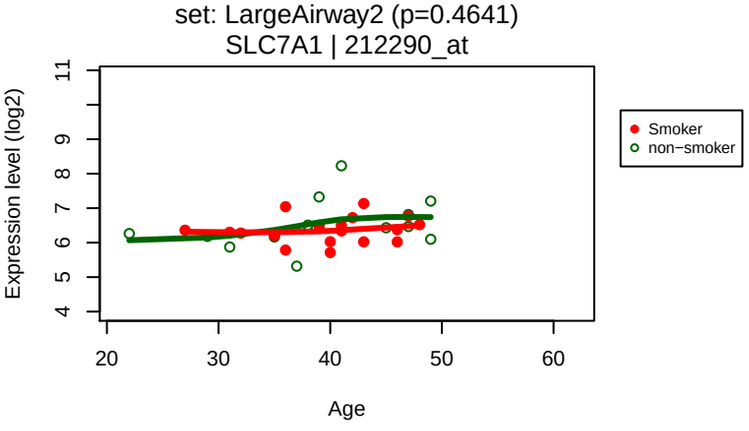

Gene expression ratio (smokers/nonsmokers)
gene: SLC7A1 | 212290_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0472
p=0.723
p=0.2922
p=0.057
p=0.4641
p=0.7666
p=0.2833
p=0.2722
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

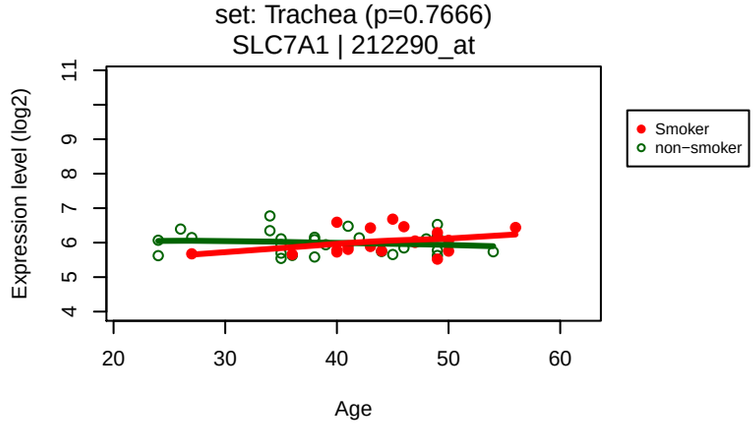

Gene expression ratio in Males
gene: SLC7A1 | 212290_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4693
p=0.5537
p=0.6983
p=0.4379
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SLC7A1 | 212290_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6379
p=0.134
p=0.1175
p=0.4436
SmallAirway2
LargeAirway2
Trachea
Macrophage2

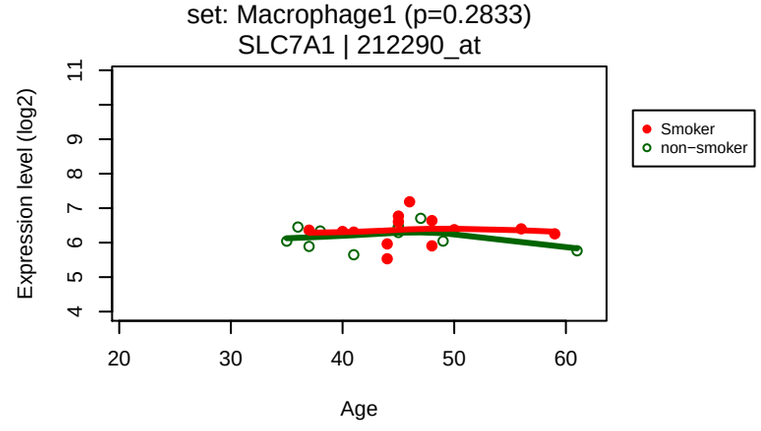

set: SmallAirway1 (p=0.0472)
SLC7A1 | 212290_at
Expression level (log2)
Age
Smoker
non−smoker

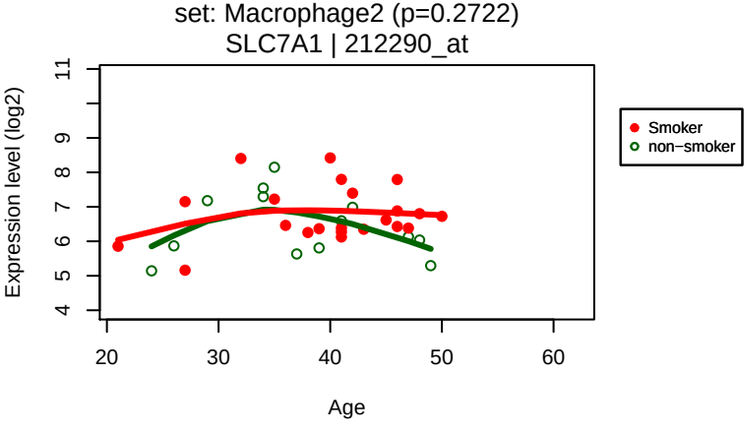

set: SmallAirway2 (p=0.723)
SLC7A1 | 212290_at
Expression level (log2)
Age
Smoker
non−smoker

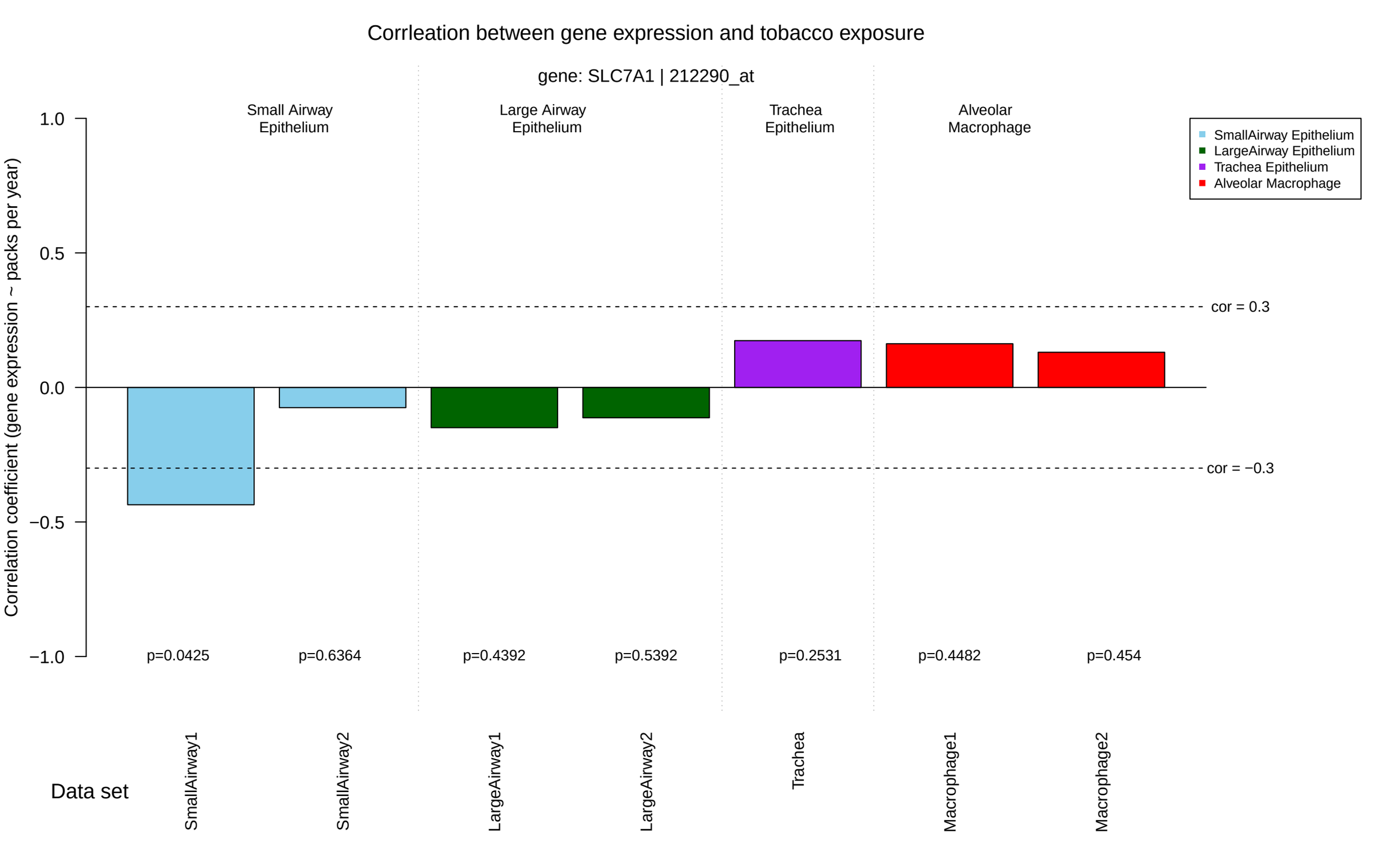

set: LargeAirway1 (p=0.057)
SLC7A1 | 212290_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4641)
SLC7A1 | 212290_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7666)
SLC7A1 | 212290_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2833)
SLC7A1 | 212290_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2722)
SLC7A1 | 212290_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SLC7A1 | 212290_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0425
p=0.6364
p=0.4392
p=0.5392
p=0.2531
p=0.4482
p=0.454
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2